MODULE 1 · LECTURE 17

Agriculture Expert Systems: Soil Information Systems for Farm Decision Support

How expert systems capture specialists' knowledge as rules and reason with it — knowledge bases, inference engines, chaining and certainty factors — and how soil information systems supply the soil data on which farm advice depends.

Learning objectives

After this lecture you should be able to:

1. Define an expert system and describe its components and the people who build it.
2. Represent agricultural knowledge as production rules, frames and semantic networks.
3. Explain forward and backward chaining and apply them to a farm problem.
4. Combine evidence with certainty factors.
5. Describe agricultural expert systems and their limitations.
6. Explain the structure and data sources of a soil information system.
7. Assign land capability classes and use soil information for farm decisions.

Scope: Lecture 16 introduced decision support systems; expert systems are knowledge-driven DSS. Soil data were used in Lectures 6 (Soil Health Card database), 13 (STCR fertiliser equations, water-holding capacity) and 15 (GIS). Artificial intelligence is introduced formally in Lecture 19; data-driven AI in Lectures 23–28.

1. What is an expert system?

An **expert system** is a computer program that uses a body of specialist knowledge and a reasoning procedure to solve problems that would normally need a human expert. Expert systems are among the earliest practical applications of artificial intelligence; they aim to perform at the level of human experts in a narrow field (Lemmon, 1986).

Agriculture needs them because specialists are few and farmers often need advice at the moment a problem appears. An expert system acts as a "virtual expert" that gives personalised advice to many farmers, drawing on several knowledge sources (Ravisankar et al., 2023).

Feature	Conventional program	Expert system
What it encodes	A fixed algorithm	Knowledge (facts and rules), separate from the reasoning procedure
How it is changed	Rewriting code	Adding or editing rules in the knowledge base
Explanation	Usually none	Can explain why it asks a question and how it reached a conclusion
Uncertain or incomplete data	Often fails	Can reason with certainty factors or fuzzy values

2. Components of an expert system

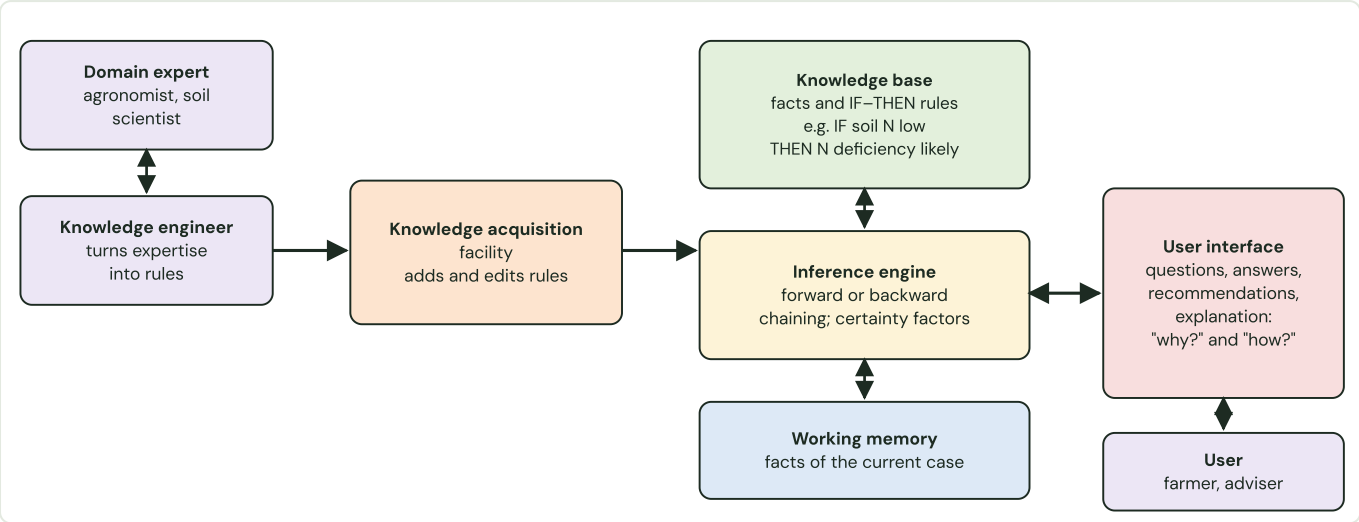


Figure 1. Components of an expert system and the people who build and use it. Original diagram.

Component	Role
Knowledge base	Holds the domain knowledge — facts and IF–THEN rules about soils, crops, pests, practices
Inference engine	Applies the rules to the facts of the case to reach conclusions (Section 4)
Working memory	Stores the facts of the current case — the farmer's answers and the conclusions reached so far
Explanation facility	Answers "why?" (why a question is asked) and "how?" (how a conclusion was reached)
Knowledge acquisition facility	Lets experts add, test and update rules
User interface	Dialogue with the user — questions, pictures, menus, recommendations

Building an expert system needs a **domain expert**, who has the knowledge, and a **knowledge engineer**, who elicits it through interviews and case studies and encodes it as rules. Collecting and coding knowledge is slow — the "knowledge acquisition bottleneck".

3. Representing knowledge

Method	Description	Agricultural example
Production rules	IF condition(s) THEN conclusion or action	IF older leaves are yellow AND soil N is low THEN nitrogen deficiency is likely
Frames	Structured records with slots and values, inheriting from more general frames	Frame "Wheat": sowing window, seed rate, critical irrigation stages, common pests
Semantic networks	Nodes (concepts) linked by labelled relations	Aphid —attacks→ mustard; ladybird beetle — eats→ aphid
Decision trees and tables	Branching questions or condition–action tables	Symptom-based diagnosis keys

4. Inference: forward and backward chaining

The inference engine links rules together in a chain (Russell & Norvig, 2020):

- **Forward chaining** (data-driven) starts from the known facts, fires every rule whose conditions are satisfied, adds the conclusions to working memory, and repeats until nothing new can be concluded. It suits monitoring and advisory tasks where data arrive first.
- **Backward chaining** (goal-driven) starts from a hypothesis or goal and works backwards to find rules that would prove it, asking only for the facts it needs. It suits diagnosis, where the system tests one possible cause at a time.

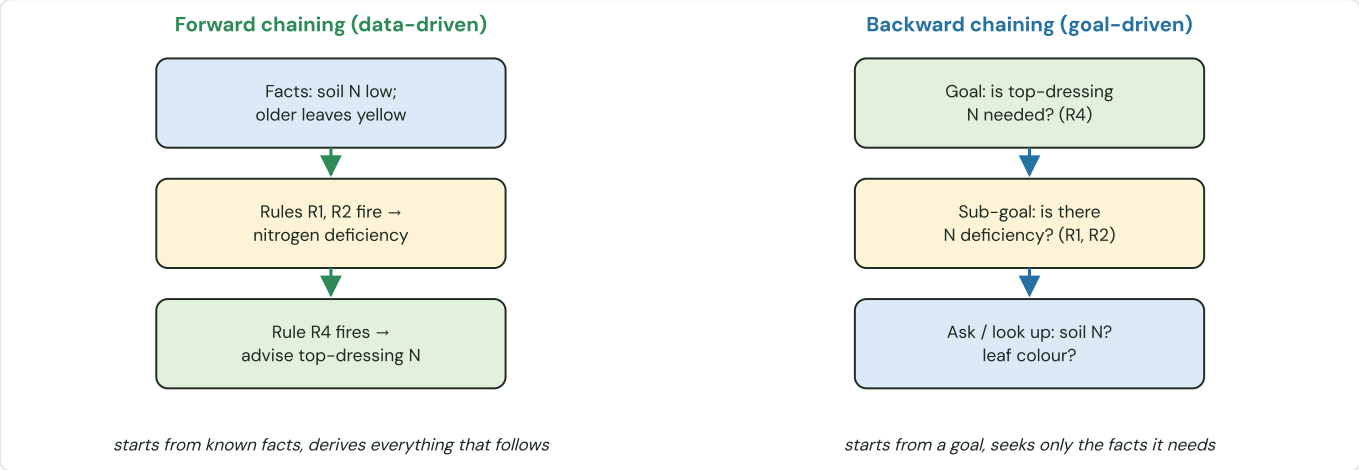


Figure 2. Forward and backward chaining on the same rules. Original diagram.

Worked example 1 — chaining with five rules (illustrative rules)
R1: IF soil N low THEN nitrogen deficiency · R2: IF older leaves yellow THEN nitrogen deficiency · R3: IF pH above 8.5 THEN sodic-soil risk · R4: IF nitrogen deficiency THEN advise top-dressing N · R5: IF sodic-soil risk THEN advise testing exchangeable sodium and considering gypsum.
Facts: soil test N = 210 kg/ha (below the 280 kg/ha "low" limit used in Lecture 6), pH = 8.8, older leaves yellow.
Forward chaining: R1, R2 and R3 fire on the facts → nitrogen deficiency and sodic-soil risk are added → R4 and R5 then fire → two pieces of advice.
Backward chaining for the goal "advise top-dressing N?": R4 needs "nitrogen deficiency" → R1 needs "soil N low" (look up the soil test: yes) → goal proved. R3 and R5 are never examined, because they are not needed for this goal.

5. Reasoning under uncertainty: certainty factors

Expert knowledge is rarely certain. The MYCIN expert system introduced **certainty factors (CF)** from -1 (definitely false) to +1 (definitely true). Each rule carries a CF given by the expert; the CF of a conclusion is the rule's CF multiplied by the certainty of its condition (the weakest condition, for AND). When two rules support the same conclusion, the new evidence is applied to the belief that still remains (Buchanan & Shortliffe, 1984):

$$\mathrm{CF}(\text{combined}) = \mathrm{CF}_1 + \mathrm{CF}_2 \times (1 - \mathrm{CF}_1) \qquad (\text{both positive}) \qquad \mathrm{CF}(\text{conclusion}) = \mathrm{CF}(\text{rule}) \times \mathrm{CF}(\text{condition})$$

Worked example 2 — combining evidence (CF values illustrative)
R1 gives nitrogen deficiency CF 0.6; R2 gives CF 0.5. Combined: 0.6 + 0.5 × (1 − 0.6) = 0.6 + 0.2 = 0.80.
R4 (CF 0.9) then gives the advice to top-dress N a certainty of 0.9 × 0.80 = 0.72.
A third, weaker piece of evidence with CF 0.4 would raise the belief to 0.80 + 0.4 × 0.20 = 0.88 — each extra item adds less, and the combined CF never exceeds 1.

A working rule-based expert system in Python (Lecture 8)

```
# A tiny rule-based expert system: forward chaining with certainty factors and explanations
RULES = [  # (rule, IF all these facts, THEN this conclusion, certainty factor of the rule)
    ("R1", ["soil N low"], "nitrogen deficiency", 0.6),
    ("R2", ["older leaves yellow"], "nitrogen deficiency", 0.5),
    ("R3", ["pH above 8.5"], "sodic soil risk", 0.7),
    ("R4", ["nitrogen deficiency"], "ADVICE: top-dress nitrogen now", 0.9),
    ("R5", ["sodic soil risk"], "ADVICE: test exchangeable sodium; consider gypsum", 0.8),
]

def combine(cf_old, cf_new):                 # MYCIN rule for two positive certainty factors
    return cf_old + cf_new * (1 - cf_old)

def forward_chain(facts):
    cf, because, fired = dict(facts), {}, set()
    changed = True
    while changed:                           # keep firing rules until nothing new is learned
        changed = False
        for name, conds, concl, rule_cf in RULES:
            if name not in fired and all(c in cf for c in conds):
                strength = min(cf[c] for c in conds) * rule_cf
                cf[concl] = combine(cf.get(concl, 0.0), strength)
                because.setdefault(concl, []).append(name + " <- " + " and ".join(conds))
                fired.add(name)
                changed = True
    return cf, because

# Facts from the soil information system and the farmer's observation
soil = {"N_kg_per_ha": 210, "pH": 8.8}
facts = {}
if soil["N_kg_per_ha"] < 280: facts["soil N low"] = 1.0
if soil["pH"] > 8.5:          facts["pH above 8.5"] = 1.0
facts["older leaves yellow"] = 1.0

cf, because = forward_chain(facts)
for concl in cf:
    if concl not in facts:
        print(f"{concl}: CF = {cf[concl]:.2f}   (because {'; '.join(because[concl])})")

Output:
nitrogen deficiency: CF = 0.80   (because R1 <- soil N low; R2 <- older leaves yellow)
sodic soil risk: CF = 0.70   (because R3 <- pH above 8.5)
ADVICE: top-dress nitrogen now: CF = 0.72   (because R4 <- nitrogen deficiency)
ADVICE: test exchangeable sodium; consider gypsum: CF = 0.56   (because R5 <- sodic soil risk)
```

The program was run and the output shown is its actual output. Its rules and CF values are illustrative, not agronomic recommendations. Fuzzy logic, which uses degrees of membership such as "moderately acidic", is another way of handling uncertainty.

6. Agricultural expert systems

System	Description
COMAX (USA, 1980s)	An expert system for cotton crop management whose knowledge base combined a cotton-plant simulation program with IF–THEN rules and an inference engine. It chose strategies for irrigation, fertiliser, defoliants and boll openers; field sensors reported weather automatically, and it re-evaluated its advice daily (Lemmon, 1986)
AgriDaksh (ICAR–IASRI, India)	An online tool for building crop expert systems, used for example to build Tobacco Agridaksh at ICAR-Central Tobacco Research Institute, which gives farmers production and protection advice (Ravisankar et al., 2023)
Diagnosis advisors	Pest, disease and nutrient-deficiency identification from symptoms (now often combined with image-based AI, Lectures 14 and 25)
Fertiliser and irrigation advisors	Rules that turn soil and crop data into doses and schedules (Lecture 13)

Strengths	Limitations
Make scarce expertise available anytime, anywhere	Knowledge acquisition is slow and costly
Consistent advice that can be explained	Work only within a narrow domain; no common sense
Preserve the knowledge of retiring experts	Rules must be updated as varieties, pests and prices change
Train students and extension staff	Cannot learn by themselves — unlike machine learning (Lectures 19–28)

7. Soil information systems

A **soil information system (SIS)** stores soil data in a database linked to a GIS (Lectures 6 and 15) and adds rules and models that interpret the data for land-use and farm decisions. Soil information is basic to almost every recommendation: crop choice, fertiliser dose, irrigation design and conservation measures.

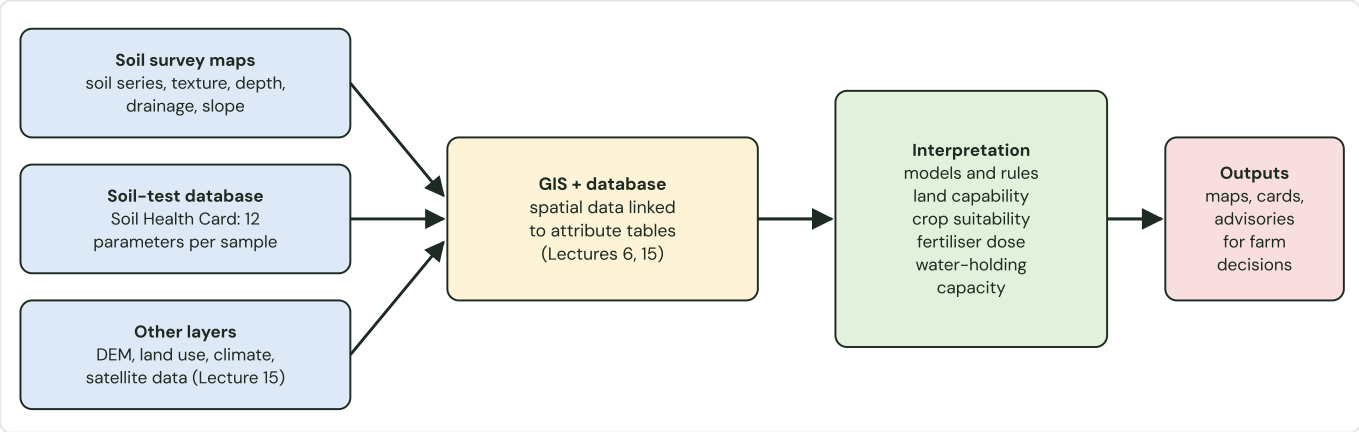


Figure 3. Structure of a soil information system. Original diagram.

7.1 Sources of soil information

Source	What it provides
Soil survey and land resource inventory	Soil maps showing soil series and their properties — depth, texture, drainage, slope, erosion. In India, the ICAR-National Bureau of Soil Survey and Land Use Planning (NBSS&LUP) prepares soil resource maps for states and districts and detailed land resource inventories for blocks and watersheds
Soil-test database	Soil Health Card results for 12 parameters — N, P, K, S, Zn, Fe, Cu, Mn, B, pH, EC and organic carbon — with fertiliser recommendations (Press Information Bureau [PIB], 2025)
Global digital soil maps	SoilGrids 2.0 maps soil properties for the whole globe at 250 m resolution, using machine learning on about 240,000 soil observations and over 400 environmental covariates, with mapped uncertainty (Poggio et al., 2021)
Pedotransfer functions	Equations that estimate hard-to-measure properties, such as field capacity and wilting point, from texture and organic matter (Saxton & Rawls, 2006) — giving the TAW used in irrigation scheduling (Lecture 13)
Sensors and remote sensing	Soil-moisture sensors (Lecture 12) and satellite indicators of soil and crop condition (Lecture 15)

7.2 Land capability classification

Land capability classification groups soils by the degree of limitation they place on use and the risk of damage under cultivation. In the widely used USDA system there are eight classes; risks and limitations increase from class I to class VIII. Subclasses show the main kind of limitation: **e** — erosion and runoff, **w** — excess water, **s** — root-zone limitations, **c** — climate (Klingebiel & Montgomery, 1961).

Class	Limitations	Use
I	Few limitations	Suited to a wide range of crops
II	Some limitations that reduce the choice of crops or need moderate conservation	Cultivation with careful management
III	Severe limitations that reduce the choice of crops or need special conservation	Cultivation with special practices
IV	Very severe limitations	Cultivation of a few crops with very careful management
V	Little or no erosion hazard, but other limitations impractical to remove (e.g. frequent flooding)	Generally not suited to cultivation; pasture, range, forest
VI	Severe limitations that make the land generally unsuited to cultivation	Pasture, range, forest, wildlife
VII	Very severe limitations	Grazing, forest, wildlife
VIII	Limitations that preclude commercial plant production	Recreation, wildlife, water supply, aesthetic use

Worked example 3 — land capability with a simple rule set (thresholds illustrative, for teaching only)
Rules: slope ≤ 1% and depth > 90 cm and well drained → class I; slope 1–3% or depth 50–90 cm → class II; slope 3–8% or depth 25–50 cm → class III; slope 8–15% → class IV. The final class is set by the most limiting factor, and the subclass names that factor.
Plot A: slope 0.5%, depth 120 cm, well drained → all factors class I → **class I**.
Plot B: slope 5%, depth 100 cm, well drained → slope gives class III, depth class I → most limiting = slope → **class IIIe** (erosion hazard).
Plot C: slope 2%, depth 40 cm → slope class II, depth class III → **class IIIs** (shallow root zone).
An SIS applies such rules to every soil polygon and draws a land-capability map in seconds.

8. Expert systems and soil information working together

1. The farmer selects a plot on a phone or the extension worker enters its location.
2. The SIS retrieves the soil series and the latest Soil Health Card values for that plot.
3. The expert system's rules interpret the soil data with the crop and season chosen — suitability, fertiliser dose (e.g. STCR equations, Lecture 13), irrigation interval from water-holding capacity.
4. The farmer's observations (leaf colour, pests) are added; the system updates its conclusions and certainty.
5. Recommendations are delivered with an explanation — why the advice is given — and stored for follow-up.

Keeping soil-based advice reliable: use recent, geo-referenced soil tests; record the sampling depth and laboratory method; state units (kg/ha, %); review rules with soil scientists every season; and show farmers the reasons for each recommendation.

9. Key takeaways

- An expert system reasons with specialists' knowledge to solve problems in a narrow field; knowledge is kept separate from the reasoning procedure.
- Its components are the knowledge base, inference engine, working memory, explanation and knowledge-acquisition facilities, and user interface.
- Knowledge is represented by production rules, frames, semantic networks and decision trees.
- Forward chaining reasons from facts to conclusions; backward chaining from a goal to the facts needed.
- Certainty factors combine evidence: $CF = CF_1 + CF_2(1 - CF_1)$ for two positive values.
- COMAX and AgriDaksh-based systems show expert systems in crop management and advisory work; their weakness is the knowledge-acquisition bottleneck.
- A soil information system links soil maps, soil tests (12 Soil Health Card parameters) and global data (SoilGrids) in a GIS, with rules for interpretation.
- Land capability classes I–VIII (subclasses e, w, s, c) express increasing limitation; the most limiting factor decides the class.

10. Examination questions

Short-answer questions (2 marks)

Q1. Define an expert system.

A computer program that uses specialist knowledge and a reasoning procedure to solve problems that normally need a human expert.

Q2. What are the roles of the domain expert and the knowledge engineer?

The domain expert supplies the knowledge; the knowledge engineer elicits it and encodes it as rules.

Q3. What does the inference engine do?

Applies the rules in the knowledge base to the facts of a case to reach conclusions.

Q4. Distinguish forward and backward chaining.

Forward chaining goes from facts to conclusions; backward chaining starts from a goal and seeks the facts that would prove it.

Q5. Give an example of a production rule in agriculture.

IF older leaves are yellow AND soil N is low THEN nitrogen deficiency is likely.

Q6. Combine certainty factors of 0.6 and 0.5 for the same conclusion.

0.6 + 0.5 × (1 − 0.6) = 0.80.

Q7. What was COMAX?

An expert system for cotton crop management that combined a crop simulation model with rules and an inference engine to advise on irrigation, fertiliser and defoliation.

Q8. Define a soil information system.

A database linked to a GIS that stores soil data and interprets them with rules and models for land-use and farm decisions.

Q9. What is SoilGrids?

A global digital soil map of soil properties at 250 m resolution produced with machine learning.

Q10. What do the subclasses e, w, s and c mean in land capability classification?

Erosion and runoff, excess water, root-zone limitations, and climatic limitations.

Long-answer questions (5–10 marks) — answer outlines

Q1. Describe the components of an expert system with a diagram.

Figure 1 and table; people involved; comparison with conventional programs.

Q2. Explain the methods of knowledge representation used in expert systems.

Rules, frames, semantic networks, decision trees — with agricultural examples.

Q3. Explain forward and backward chaining with an agricultural example.

Figure 2 and worked example 1.

Q4. Explain certainty factors and how evidence is combined. Illustrate with a program.

CF range and rules; combining formula; worked example 2; the Python program and its output.

Q5. Describe agricultural expert systems and discuss their strengths and limitations.

COMAX, AgriDaksh, diagnosis and fertiliser advisors; strengths and limitations table.

Q6. Describe the structure and data sources of a soil information system and explain land capability classification.

Figure 3; sources table (survey, SHC, SoilGrids, pedotransfer functions, sensors); classes I–VIII and subclasses; worked example 3; Section 8 workflow.

11. Glossary

Term	Meaning
Backward chaining	Goal-driven reasoning from a hypothesis back to the facts
Certainty factor	A number from −1 to +1 expressing belief in a fact or rule
Domain expert	The specialist whose knowledge is captured
Explanation facility	Part of an expert system that explains its questions and conclusions
Forward chaining	Data-driven reasoning from facts to conclusions
Frame	A structured record of slots and values describing a concept
Inference engine	The reasoning component that applies rules to facts
Knowledge base	The store of facts and rules
Knowledge engineer	The person who elicits and encodes expert knowledge
Land capability class	One of eight classes (I–VIII) of increasing limitation for use
Pedotransfer function	Equation estimating soil properties from easily measured ones
Production rule	An IF–THEN statement of knowledge
Semantic network	A network of concepts linked by labelled relations
SIS	Soil information system
Soil series	A group of soils with similar profiles, used as a mapping unit
Working memory	Facts of the current case held during reasoning

References

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All diagrams and the program are original. Rules, certainty factors and land-capability thresholds in the examples are illustrative teaching values, not official criteria or recommendations.

Practice MCQs

Q1. An expert system mainly encodes

(a) a fixed sequence of calculations
(b) specialist knowledge as facts and rules
(c) only images
(d) only a database of prices

Q2. The component that applies rules to the facts of a case is the

(a) knowledge base
(b) inference engine
(c) user interface
(d) working memory

Q3. The person who elicits knowledge from experts and encodes it as rules is the

(a) domain expert
(b) end user
(c) knowledge engineer
(d) database administrator

Q4. Reasoning that starts from known facts and derives conclusions is

(a) backward chaining
(b) forward chaining
(c) normalisation
(d) compilation

Q5. Backward chaining is especially suited to

(a) diagnosis of a suspected cause
(b) storing soil maps
(c) printing reports
(d) drawing charts

Q6. Two rules give the same conclusion CF 0.6 and 0.5. The combined CF is

(a) 0.30
(b) 0.55
(c) 0.80
(d) 1.10

Q7. Which part of an expert system answers "how did you reach this conclusion?"

(a) explanation facility
(b) knowledge acquisition facility
(c) compiler
(d) GIS

Q8. COMAX was an expert system for

(a) rice irrigation
(b) cotton crop management
(c) dairy feeding
(d) wheat marketing

Q9. A soil information system links soil data to

(a) a GIS and a database
(b) a word processor
(c) only paper maps
(d) a spreadsheet chart

Q10. SoilGrids 2.0 maps global soil properties at a resolution of

(a) 10 m
(b) 250 m
(c) 1 km
(d) 10 km

Q11. In land capability classification, limitations increase

(a) from class VIII to class I
(b) from class I to class VIII
(c) only in class IV
(d) randomly

Q12. In land capability subclasses, "w" stands for

(a) wind
(b) excess water
(c) weeds
(d) wilting

Answers

Q1 — (b) specialist knowledge as facts and rules
Q2 — (b) inference engine
Q3 — (c) knowledge engineer
Q4 — (b) forward chaining
Q5 — (a) diagnosis of a suspected cause

Q6 — (c) 0.80
Q7 — (a) explanation facility
Q8 — (b) cotton crop management
Q9 — (a) a GIS and a database
Q10 — (b) 250 m
Q11 — (b) from class I to class VIII
Q12 — (b) excess water